

ACR36521

## Boron tribromide

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:**  
**Product Description:** 三溴化硼  
Boron tribromide

**Cat No. :** 365210000; 365210250; 365211000; 365215000  
**Synonyms** Tribromoboron; Boron bromide.; Tribromoborane  
**CAS No** 10294-33-4  
**Molecular Formula** B Br<sub>3</sub>

**Supplier** **UK entity/business name**  
Fisher Scientific UK  
Bishop Meadow Road,  
Loughborough, Leicestershire LE11 5RG, United Kingdom

**EU entity/business name**  
Thermo Fisher Scientific  
Janssen Pharmaceuticaaan 3a, 2440 Geel, Belgium

**Emergency Telephone Number** For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Liquid

**Appearance**  
Amber

**Odor**  
pungent

#### Emergency Overview

Causes severe skin burns and eye damage. Fatal if swallowed. Fatal if inhaled. Reacts violently with water.

#### Classification of the substance or mixture

|                                    |              |
|------------------------------------|--------------|
| Acute Oral Toxicity                | Category 2   |
| Acute Inhalation Toxicity - Vapors | Category 2   |
| Skin Corrosion/Irritation          | Category 1 A |
| Serious Eye Damage/Eye Irritation  | Category 1   |

#### Label Elements



**Signal Word****Danger****Hazard Statements**

H314 - Causes severe skin burns and eye damage

H300 + H330 - Fatal if swallowed or if inhaled

**Precautionary Statements****Prevention**

P264 - Wash face, hands and any exposed skin thoroughly after handling

P270 - Do not eat, drink or smoke when using this product

P271 - Use only outdoors or in a well-ventilated area

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P284 - Wear respiratory protection

**Response**

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor

P330 - Rinse mouth

P331 - Do NOT induce vomiting

P362 + P364 - Take off contaminated clothing and wash it before reuse

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

P405 - Store locked up

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

Reacts violently with water.

**Health Hazards**

Corrosive. Causes skin and eye burns. Fatal if inhaled. Fatal if swallowed.

**Environmental hazards**

Reacts violently with water. Is not likely mobile in the environment. Spillage unlikely to penetrate soil.

This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component        | CAS No     | Weight % |
|------------------|------------|----------|
| Boron tribromide | 10294-33-4 | >95      |

**SECTION 4. FIRST AID MEASURES****General Advice**

Immediate medical attention is required. Show this safety data sheet to the doctor in attendance.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

**Inhalation**

Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Remove to fresh air. Immediate medical attention is required. If not breathing, give artificial respiration.

**Ingestion**

Do NOT induce vomiting. Call a physician or poison control center immediately.

**Most important symptoms and effects**

Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

**Extinguishing media which must not be used for safety reasons**

Water.

**Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Reacts violently with water.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Use personal protective equipment as required. Ensure adequate ventilation.

**Environmental Precautions**

Should not be released into the environment. See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Do not expose spill to water.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Do not breathe mist/vapors/spray. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not ingest. If swallowed then seek immediate medical assistance. Do not allow contact with water.

**Storage**

Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area. Keep away from water or moist air.

**Specific Use(s)**

Use in laboratories

## Boron tribromide

## SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

## Control Parameters

| Component        | China | Taiwan | Thailand       | Hong Kong                                       |
|------------------|-------|--------|----------------|-------------------------------------------------|
| Boron tribromide | -     | -      | Ceiling: 1 ppm | Ceiling: 1 ppm<br>Ceiling: 10 mg/m <sup>3</sup> |

| Component        | ACGIH TLV        | OSHA PEL                                                            | NIOSH                                           | The United Kingdom                                      | European Union |
|------------------|------------------|---------------------------------------------------------------------|-------------------------------------------------|---------------------------------------------------------|----------------|
| Boron tribromide | Ceiling: 0.7 ppm | (Vacated) Ceiling: 1 ppm<br>(Vacated) Ceiling: 10 mg/m <sup>3</sup> | Ceiling: 1 ppm<br>Ceiling: 10 mg/m <sup>3</sup> | STEL: 1 ppm 15 min<br>STEL: 10 mg/m <sup>3</sup> 15 min |                |

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

## Exposure Controls

## Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

## Personal protective equipment

## Eye Protection

If splashes are likely to occur: Goggles Face protection shield (European standard - EN 166)

## Hand Protection

Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Butyl rubber   | > 120 minutes     | 0.7 mm          | EN 374      | (minimum requirement) |
| Viton (R)      | > 480 minutes     | 0.7 mm          |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

## Skin and body protection

Long sleeved clothing

## Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

## Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced

**Recommended Filter type:** Acid gases filter Type E Yellow conforming to EN14387

## Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141

When RPE is used a face piece Fit Test should be conducted

## Boron tribromide

|                                 |                                                                        |
|---------------------------------|------------------------------------------------------------------------|
| Hygiene Measures                | Handle in accordance with good industrial hygiene and safety practice. |
| Environmental exposure controls | No information available.                                              |

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                         |                             |                                   |
|-----------------------------------------|-----------------------------|-----------------------------------|
| Appearance                              | Amber                       |                                   |
| Physical State                          | Liquid                      |                                   |
| Odor                                    | pungent                     |                                   |
| Odor Threshold                          | No data available           |                                   |
| pH                                      | No information available    |                                   |
| Melting Point/Range                     | -46 °C / -50.8 °F           |                                   |
| Softening Point                         | No data available           |                                   |
| Boiling Point/Range                     | 91.2 °C / 196.2 °F          | @ 760 mmHg                        |
| Flash Point                             | No information available    | Method - No information available |
| Evaporation Rate                        | No data available           |                                   |
| Flammability (solid,gas)                | Not applicable              | Liquid                            |
| Explosion Limits                        | No data available           |                                   |
| Vapor Pressure                          | 40 mmHg @ 14 °C             |                                   |
| Vapor Density                           | 8.6                         | (Air = 1.0)                       |
| Specific Gravity / Density              | 2.650                       |                                   |
| Bulk Density                            | Not applicable              | Liquid                            |
| Water Solubility                        | Reacts violently with water |                                   |
| Solubility in other solvents            | No information available    |                                   |
| Partition Coefficient (n-octanol/water) | No data available           |                                   |
| Autoignition Temperature                | No data available           |                                   |
| Decomposition Temperature               | No data available           |                                   |
| Viscosity                               | 0.731 mPas @ 24°C           | Dynamic                           |
| Explosive Properties                    | No information available    |                                   |
| Oxidizing Properties                    | No information available    |                                   |
| Molecular Formula                       | B Br <sub>3</sub>           |                                   |
| Molecular Weight                        | 250.52                      |                                   |

## SECTION 10. STABILITY AND REACTIVITY

|                                  |                                                                                                              |
|----------------------------------|--------------------------------------------------------------------------------------------------------------|
| Stability                        | Stable under normal conditions.                                                                              |
| Hazardous Reactions              | Reacts violently with water.                                                                                 |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                                     |
| Conditions to Avoid              | Incompatible products. Excess heat. Exposure to moist air or water. Exposure to moisture.                    |
| Materials to avoid               | Acids. Water. Alcohols. Metals.                                                                              |
| Hazardous Decomposition Products | Hydrogen halides. Oxides of boron. Thermal decomposition can lead to release of irritating gases and vapors. |

## SECTION 11. TOXICOLOGICAL INFORMATION

|                     |                                           |
|---------------------|-------------------------------------------|
| Product Information | Very toxic by inhalation and if swallowed |
|---------------------|-------------------------------------------|

## (a) acute toxicity;

| Component        | LD50 Oral | LD50 Dermal | LC50 Inhalation                                       |
|------------------|-----------|-------------|-------------------------------------------------------|
| Boron tribromide |           |             | LC50 = 2858 ppm/1H (rat)<br>LC50 = 814 ppm/1H (mouse) |

|                                            |                                                                                                                                                                                                                                                            |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (b) skin corrosion/irritation;             | Category 1 A                                                                                                                                                                                                                                               |
| (c) serious eye damage/irritation;         | Category 1                                                                                                                                                                                                                                                 |
| (d) respiratory or skin sensitization;     |                                                                                                                                                                                                                                                            |
| Respiratory                                | No data available                                                                                                                                                                                                                                          |
| Skin                                       | No data available                                                                                                                                                                                                                                          |
| (e) germ cell mutagenicity;                | No data available                                                                                                                                                                                                                                          |
| (f) carcinogenicity;                       | No data available                                                                                                                                                                                                                                          |
|                                            | There are no known carcinogenic chemicals in this product                                                                                                                                                                                                  |
| (g) reproductive toxicity;                 | No data available                                                                                                                                                                                                                                          |
| (h) STOT-single exposure;                  | No data available                                                                                                                                                                                                                                          |
| (i) STOT-repeated exposure;                | No data available                                                                                                                                                                                                                                          |
| Target Organs                              | No information available.                                                                                                                                                                                                                                  |
| (j) aspiration hazard;                     | No data available                                                                                                                                                                                                                                          |
| Symptoms / effects, both acute and delayed | Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |

**SECTION 12. ECOLOGICAL INFORMATION**

|                                 |                                                                                                                                                             |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ecotoxicity effects             | Reacts with water so no ecotoxicity data for the substance is available. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms. |
| Persistence and Degradability   |                                                                                                                                                             |
| Persistence                     | Persistence is unlikely, based on information available.                                                                                                    |
| Degradability                   | Not relevant for inorganic substances, Reacts with water.                                                                                                   |
| Bioaccumulative Potential       | Product does not bioaccumulate due to reaction with water                                                                                                   |
| Mobility in soil                | Spillage unlikely to penetrate soil. Is not likely mobile in the environment                                                                                |
| Endocrine Disruptor Information | This product does not contain any known or suspected endocrine disruptors                                                                                   |
| Persistent Organic Pollutant    | This product does not contain any known or suspected substance                                                                                              |
| Ozone Depletion Potential       | This product does not contain any known or suspected substance                                                                                              |

**SECTION 13. DISPOSAL CONSIDERATIONS**

|                            |                                                                                         |
|----------------------------|-----------------------------------------------------------------------------------------|
| Waste from Residues/Unused | Waste is classified as hazardous. Dispose of in accordance with the European Directives |
|----------------------------|-----------------------------------------------------------------------------------------|

## Boron tribromide

|                        |                                                                                                                                                                                                               |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Products               | on waste and hazardous waste. Dispose of in accordance with local regulations.                                                                                                                                |
| Contaminated Packaging | Dispose of this container to hazardous or special waste collection point.                                                                                                                                     |
| Other Information      | Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not flush to sewer. Large amounts will affect pH and harm aquatic organisms. |

## SECTION 14. TRANSPORT INFORMATION

## Road and Rail Transport

|                      |                  |
|----------------------|------------------|
| UN-No                | UN2692           |
| Proper Shipping Name | BORON TRIBROMIDE |
| Hazard Class         | 8                |
| Packing Group        | I                |

## IMDG/IMO

|                      |                  |
|----------------------|------------------|
| UN-No                | UN2692           |
| Proper Shipping Name | BORON TRIBROMIDE |
| Hazard Class         | 8                |
| Packing Group        | I                |

## IATA

FORBIDDEN FOR IATA TRANSPORT

|                      |                                               |
|----------------------|-----------------------------------------------|
| UN-No                | UN2692                                        |
| Proper Shipping Name | BORON TRIBROMIDE FORBIDDEN FOR IATA TRANSPORT |
| Hazard Class         | 8                                             |
| Packing Group        | I                                             |

Special Precautions for User No special precautions required

## SECTION 15. REGULATORY INFORMATION

## International Inventories

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component        | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Boron tribromide | X                                                   | X                                       | X    | X     | 233-657-9 | X    | X   | X     | X    | X    | X    | KE-03538 |

## National Regulations

## SECTION 16. OTHER INFORMATION

|                  |                 |
|------------------|-----------------|
| Creation Date    | 14-May-2010     |
| Revision Date    | 07-Apr-2024     |
| Revision Summary | Not applicable. |

## Boron tribromide

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

**Legend**

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**